

Practice Quiz 09A — MATH 2055 — Fall 2026 — Solutions

Practice for Quiz 09

Well-definedness on a quotient; onto maps and inverses.

Write $[x]_5$ for the congruence class of x modulo 5.

1. Consider

$$f : \mathbf{Z}_5 \longrightarrow \mathbf{Z}_5, \quad f([x]_5) = [2x + 1]_5.$$

(a) Prove that f is well defined.

Solution Suppose $[x]_5 = [y]_5$. Then $5 \mid (x - y)$, so $x - y = 5k$ for some integer k . Therefore

$$(2x + 1) - (2y + 1) = 2(x - y) = 5(2k).$$

Thus $5 \mid ((2x + 1) - (2y + 1))$, so $[2x + 1]_5 = [2y + 1]_5$. Hence the output does not depend on the representative and f is well defined. ■

(b) Prove that f is onto and determine f^{-1} .

Solution The number 3 is an inverse of 2 modulo 5, since $2 \cdot 3 \equiv 1 \pmod{5}$. Given an arbitrary $[y]_5$, choose

$$[x]_5 = [3(y - 1)]_5.$$

Then

$$f([x]_5) = [2 \cdot 3(y - 1) + 1]_5 = [y]_5,$$

because $2 \cdot 3 \equiv 1 \pmod{5}$. Thus f is onto. The calculation also gives

$$f^{-1}([y]_5) = [3(y - 1)]_5.$$

The same representative check as in part (a), with coefficient 3, shows that this inverse rule is well defined. ■